

Chapter 2 – Motion in One Dimension

Position, velocity and speed
Acceleration
Free Fall

Constant v_x
Constant a_x
Kinematic Equations

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Kinematics and Motion

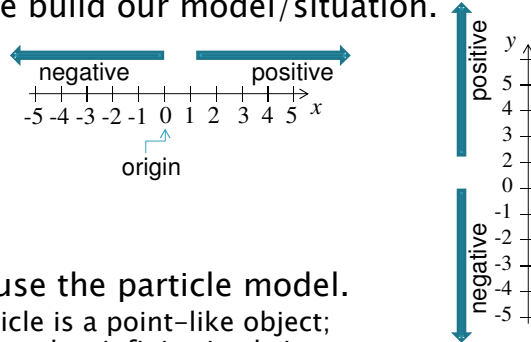
- ▶ Kinematics: describes motion not what caused or modified the motion
- ▶ For now, motion in one dimension (straight line), later in 2-D
- ▶ Also other types of motion
 - Translational (Ch 2 and 4)
 - Rotational (Ch 10, 11)
 - Vibrational (Ch 15, 16–17)

Introduction

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Co-ordinate system

- Choose a reference frame (co-ordinate system) in which we build our model/situation.



- We will use the particle model.
 - A particle is a point-like object; has mass but infinitesimal size
 - Will see how we justify this in later chapters

Introduction

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Position

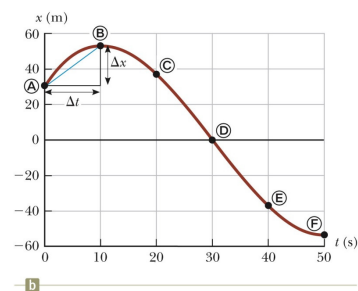
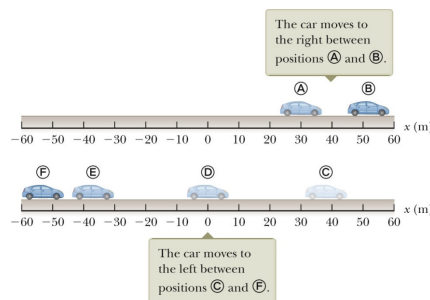
- The position of an object is usually measured with respect to the origin on the co-ordinate system at various times
- Vector
- Tabulate and graph

TABLE 2.1

Position of the Car at Various Times

Position	t (s)	x (m)
(A)	0	30
(B)	10	52
(C)	20	38
(D)	30	0
(E)	40	-37
(F)	50	-53

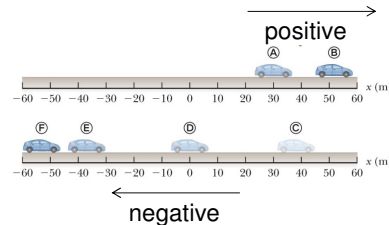
Section 2.1



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Displacement

- ▶ Displacement is defined as the change in position during some time interval.
 - Represented as Δx (Delta “x”) in 1-D motion
 - $\Delta x \equiv x_f - x_i$
 - SI units: meters (m)
- ▶ Δx is a vector quantity
 - $\therefore \Delta x$ can be positive or negative



- ▶ Displacement $\neq$ distance
 - Distance, d , is the length of a path followed by a particle.
 - Scalar quantity – always positive.

Section 2.1

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Distance vs. Displacement: Example

- ▶ A basketball player: moves from one end of the court to the other (28.7 m) and back.
- ▶ Distance covered:
 - Total path length (57.4 m)
- ▶ Displacement:
 - $\Delta x = x_f - x_i = 0$ since $x_f = x_i$



Section 2.1

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Vectors and Scalars

- ▶ Vector quantities need magnitude (size or numerical value) and direction and units to completely describe them.
 - Will use + and - signs to indicate vector directions in this chapter
- ▶ Scalar quantities are completely described by magnitude and units only.

Section 2.1

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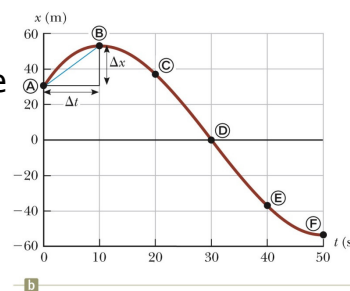
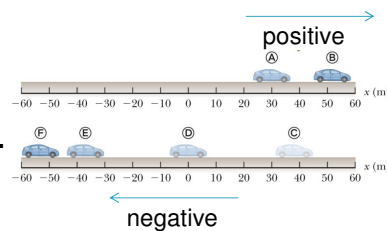
Average Velocity

- ▶ The **average velocity** is rate at which the displacement occurs.

$$v_{x,avg} \equiv \frac{\Delta x}{\Delta t} = \frac{x_f - x_i}{\Delta t}$$

- The x indicates motion along the x -axis.

- ▶ Vector quantity
- ▶ Dimensions are L/T
- ▶ The SI units: m/s
- ▶ This is also the slope of the line in the position - time graph



Section 2.1

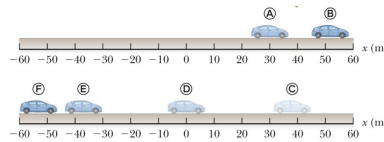
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Average Speed

- Speed is a scalar quantity.
 - Has the same units as velocity
 - Defined as **total distance** / total time:

$$v_{avg} \equiv \frac{d}{\Delta t}$$

- The speed has no direction and is always expressed as a positive number.



$$v_{x,avg} \equiv \frac{\Delta x}{\Delta t} = \frac{x_f - x_i}{\Delta t}$$

VS

$$v_{avg} \equiv \frac{d}{t}$$

Section 2.1

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Average Speed vs Average Velocity Example

- A 400 m runner runs around the track and ends at her starting point in 60 s.
 - Displacement $\Delta x = \underline{\hspace{2cm}}$
 - The average velocity $v_{x,avg} = \underline{\hspace{2cm}}$
 - Distance traveled $d = \underline{\hspace{2cm}}$
 - speed $v_{avg} = \underline{\hspace{2cm}}$
- Note: The average speed is not the magnitude of the average velocity.

Section 2.1

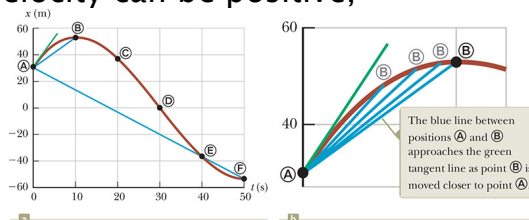
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Instantaneous Velocity

- ▶ Instantaneous velocity indicates what is happening at every point in time:

$$v_x = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt}$$

- ▶ Graphically, this represents the slope of the tangent to the x vs t graph at a specific time.
- ▶ The instantaneous velocity can be positive, negative, or zero.



Section 2.2

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Instantaneous Speed

- ▶ The instantaneous speed is the **magnitude of the instantaneous velocity**.
- ▶ The instantaneous speed is always positive and has no direction associated with it.
- ▶ Note the difference: average speed (not related to average velocity) and instantaneous speed (magnitude of instantaneous velocity)

Section 2.2

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Examples

- ▶ Consider the following equations for the position $x(t)$ of a particle in four situations (x is in m, t is in s):

- In which situation is the velocity of the particle constant?
- In which is the velocity in the negative x direction?

$$x = 3t - 2$$

$$x = -4t^2 - 2$$

$$x = \frac{2}{t^2}$$

$$x = -2$$

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Examples

- ▶ Consider the following equations for the position $x(t)$ of a particle in four situations (x is in m, t is in s):

- In which situation is the velocity of the particle constant?
- In which is the velocity in the negative x direction?

$$x = 3t - 2$$

$$v = \frac{dx}{dt} = 3$$

$$x = -4t^2 - 2$$

$$v = \frac{dx}{dt} = -8t$$

$$x = \frac{2}{t^2}$$

$$v = \frac{dx}{dt} = -\frac{4}{t^3}$$

$$x = -2$$

$$v = \frac{dx}{dt} = 0$$

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Vocabulary Note

- ▶ “Velocity” and “speed” will indicate *instantaneous* values.
- ▶ *Average* will be used when the average velocity or average speed is indicated.

Section 2.2

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Model: A Particle Under Constant Velocity

▶ Constant velocity

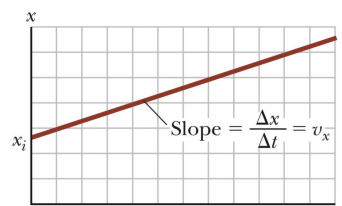
- $v_x = v_{x, avg}$
- The mathematical representation of this situation is the equation:

$$v_{x, avg} = v_x = \frac{\Delta x}{\Delta t} = \frac{x_f - x_i}{\Delta t} \quad \text{or} \quad x_f = x_i + v_x \Delta t$$

- Common practice is to let $t_i = 0$ and the equation becomes:

$$x_f = x_i + v_x t \quad (\text{for constant } v_x)$$

- ▶ Graph – motion of a particle under constant velocity (x vs t).
- ▶ Slope – constant velocity, v_x .
- ▶ The y-intercept is x_i (when $t = 0$)



Section 2.3

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Analysis model approach to problem solving.

- ▶ Please read through section 2.4, which will give you some ideas as to how to approach problems, both in Physics and in general.
- ▶ Work through Example 2.4 on page 32 which is a repeat of the Example on page 29.

Section 2.4

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Average Acceleration

- ▶ Acceleration is the rate of change of the velocity.
- ▶ Average acceleration:
$$a_{x,avg} \equiv \frac{\Delta v_x}{\Delta t} = \frac{v_{xf} - v_{xi}}{t_f - t_i}$$
- ▶ Dimensions: L/T²
- ▶ SI units: m/s²
- ▶ In one dimension, positive and negative can be used to indicate direction.

Section 2.5

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Instantaneous Acceleration

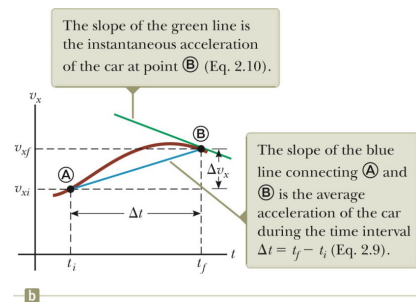
- ▶ The instantaneous acceleration

$$a_x = \lim_{\Delta t \rightarrow 0} \frac{\Delta v_x}{\Delta t} = \frac{dv_x}{dt} = \frac{d^2x}{dt^2}$$

- ▶ The slope of the velocity-time (v_x vs t) graph is the acceleration.
- ▶ The green line.
- ▶ The blue line.

+ve acceleration

-ve acceleration ("deacceleration")



Section 2.5

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Example

A particle's position on the x axis is given by $x = 4 - 27t + t^3$ with x in meters and t in seconds.

- Because position depends on t , the particle must be moving. Find the particle's velocity function $v(t)$ and acceleration function $a(t)$.
- Is there ever a time when $v = 0$?
- Describe the particle's motion for $t \geq 0$

Section 2.5

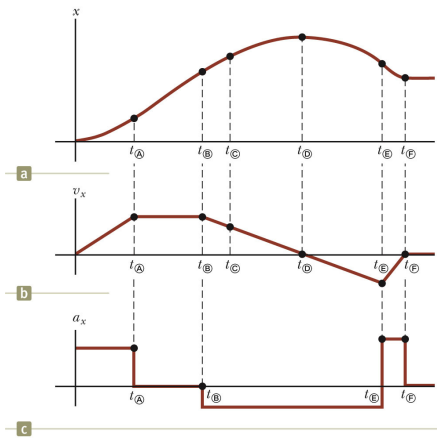
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Graphical Relationships

▶ Given the displacement–time graph (a)

▶ The velocity–time graph is obtained from the slope of (a) at every instant – (b).

▶ The acceleration–time graph is obtained from the slope of (b) at every instant – (c).



Section 2.5

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Acceleration and Velocity, Directions

- ▶ When an object's velocity and acceleration are in the same direction, the object is speeding up.
- ▶ When an object's velocity and acceleration are in the opposite direction, the object is slowing down.
- ▶ Negative acceleration does not necessarily mean the object is slowing down.
 - If the acceleration and velocity are both negative, the object is speeding up in the negative direction.
- ▶ The word *deceleration* has the connotation of slowing down.
 - This word will not be used in the text.

Section 2.6

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Analysis Model: Particle under constant acceleration

- ▶ In this model the acceleration is constant, and $a_x = a_{x,avg}$
- ▶ There is a set of equations which we will derive for this model.
- ▶ These equations can ONLY be applied when the object/particle is under uniform (constant) acceleration in one-dimension.
- ▶ You may need to use two of the equations to solve one problem.
- ▶ Many times there is more than one way to solve a problem.

Section 2.6

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Particle under constant acceleration

- ▶ Position, velocity and acceleration curves
- ▶ Special set of equations

$$a_x = a_{x,avg} = \frac{v_{x,f} - v_{x,i}}{t} \quad (1)$$

- ▶ $v_{x,i}$ is initial velocity at $t = 0$
- ▶ $v(t)$ is linear, $a(t) = \text{slope} = \text{const.}$

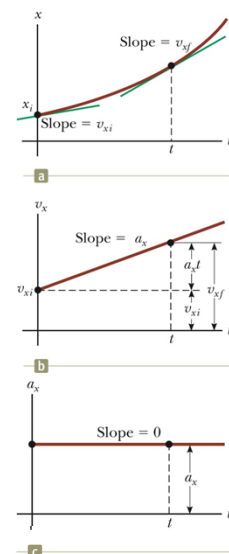
$$v_{x,avg} = \frac{x_f - x_i}{t} \quad (2)$$

- ▶ x_i is initial position at $t = 0$, $v_{x,avg}$ is average velocity between $t = 0$ and later time.

$$v_{x,avg} = \frac{1}{2}(v_{x,i} + v_{x,f}) \quad (3)$$

- ▶ (1) into (3) gives: $v_{x,avg} = v_{x,i} + \frac{1}{2}a_x t$ (4)

- ▶ (4) into (2) gives: $x_f = x_i + v_{x,i}t + \frac{1}{2}a_x t^2$ (5)



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Particle under constant acceleration

5 equations

$$v_{xf} = v_{xi} + a_x t$$

$$v_{x,avg} = \frac{1}{2}(v_{xi} + v_{xf})$$

$$v_{x,avg} = v_{xi} + \frac{1}{2}a_x t$$

$$x_f = x_i + v_{x,avg} t$$

$$x_f = x_i + v_{xi} t + \frac{1}{2}a_x t^2$$

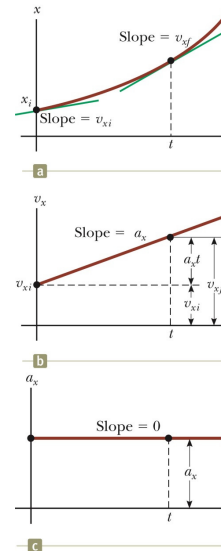
Other eqns derived from above

$$v_{xf}^2 = v_{xi}^2 + 2a_x(x_f - x_i)$$

$$x_f - x_i = \frac{1}{2}(v_{xi} + v_{xf})t$$

$$x_f - x_i = v_{xf} t - \frac{1}{2}a_x t^2$$

- Must be able to **derive ALL above equations** from first principles (i.e. Def. of a_x and v_x)



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Particle under constant acceleration Summary

TABLE 2.2 Kinematic Equations for Motion of a Particle
Under Constant Acceleration

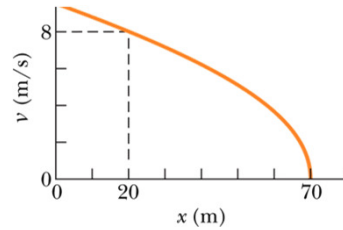
Equation Number	Equation	Information Given by Equation
2.13	$v_{xf} = v_{xi} + a_x t$	Velocity as a function of time
2.15	$x_f = x_i + \frac{1}{2}(v_{xi} + v_{xf})t$	Position as a function of velocity and time
2.16	$x_f = x_i + v_{xi} t + \frac{1}{2}a_x t^2$	Position as a function of time
2.17	$v_{xf}^2 = v_{xi}^2 + 2a_x(x_f - x_i)$	Velocity as a function of position

Note: Motion is along the x axis.

Example

The figure gives the particles velocity v versus position as it moves along an x axis with constant acceleration. What is its velocity at position $x = 0$?

? What do we know – from graph and question?



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Example

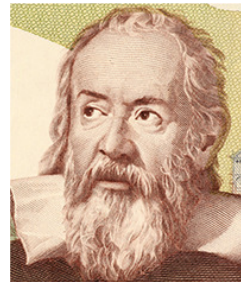
A car is slowing down at a constant rate from 75 km/h to 45 km/h over a distance of 88 m. (a) Determine the acceleration. (b) Determine the time that elapses when covering the 88 m. (c) How long will it take to bring the car to rest? (d) Over what distance will it take to come to rest?

? What do we know?

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Freely Falling Objects

- ▶ A ***freely falling object*** is any object moving freely under the influence of gravity alone.
- ▶ It does not depend upon the initial motion of the object.
 - Dropped – released from rest
 - Thrown downward
 - Thrown upward
- ▶ Galileo Galilei formulated laws of motion for objects in free fall



Georgios Kallidas/Shutterstock.com

Section 2.7

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Acceleration of Freely Falling Object

- ▶ The acceleration of an object in free fall is directed downward, regardless of the initial motion.
- ▶ The magnitude of free fall acceleration is $g = 9.80 \text{ m/s}^2$.
 - g decreases with increasing altitude
 - g varies with latitude
 - 9.80 m/s^2 is the average at the Earth's surface (We do not use 10 m/s^2 .)
 - **DO NOT USE** ~~$g = -9.8 \text{ m/s}^2$~~ . 
 - The italicized g will be used for the acceleration due to gravity.
 - Not to be confused with g for grams

Section 2.7

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Acceleration of Free Fall, cont.

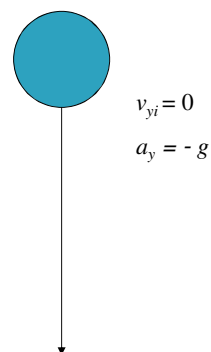
- ▶ We will neglect air resistance.
- ▶ Free fall motion is constantly accelerated motion in one dimension (downwards).
- ▶ Let upward be positive
- ▶ Use the kinematic equations
 - With $a_y = -g = -9.80 \text{ m/s}^2$
 - Note displacement is in the vertical direction (y -axis)

Section 2.7

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Free Fall – An Object Dropped

- ▶ Initial velocity is zero
- ▶ Let up be positive
- ▶ Use the kinematic equations
 - Generally use y instead of x since motion is vertical
- ▶ Acceleration is
 - $a_y = -g = -9.80 \text{ m/s}^2$

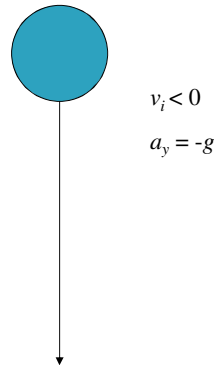


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Free Fall – An Object Thrown Downward

- ▶ $a_y = -g = -9.80 \text{ m/s}^2$
- ▶ Initial velocity $\neq 0 \text{ m/s}$
 - With upward being positive, initial velocity will be negative.

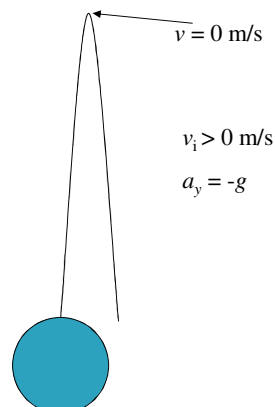


Section 2.7

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Free Fall – Object Thrown Upward

- ▶ Initial velocity is upward, so positive
- ▶ The instantaneous velocity at the maximum height is zero.
- ▶ $a_y = -g = -9.80 \text{ m/s}^2$ throughout the motion.



Section 2.7

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Thrown upward, cont.

- ▶ The motion may be **symmetrical** if the initial vertical position/height is the same as the final height.
 - Then $t_{up} = t_{down}$
 - Then $v_f = -v_i$
- ▶ The motion can also be **non-symmetrical**, when the initial and final heights of the object are not the same.
 - Break the motion into various parts.

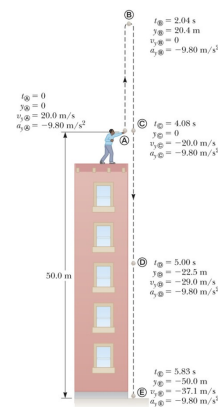
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Free Fall Example

A stone thrown from the top of a building is given an initial velocity of 20.0 m/s straight upward. The stone is launched 50 m above the ground, and the stone just misses the edge of the roof on its way down.

- a) How long does it take the stone to reach max height?
- b) Find the max height
- c) Determine the velocity of the stone when it returns to its launch height.
- d) Find the velocity and position of the stone at $t = 5.00$ s

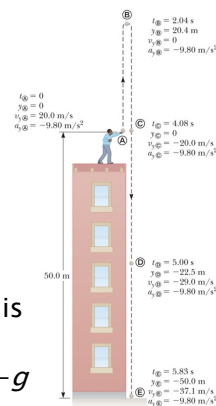


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Free Fall Example

A stone thrown from the top of a building is given an initial velocity of 20.0 m/s straight upward. The stone is launched 50 m above the ground, and the stone just misses the edge of the roof on its way down.

- ▶ Initial velocity at A is upward (+) and acceleration is $-g$ (-9.8 m/s^2).
- ▶ At B, the velocity is 0 m/s and the acceleration is $-g$ (-9.8 m/s^2).
- ▶ At C, the velocity has the same magnitude as at A, but is in the opposite direction (negative).
- ▶ At E, the displacement is -50.0 m (it ends up 50.0 m below its starting point).

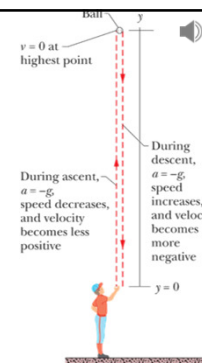


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Example

A pitcher tosses a baseball up (along a y axis) with an initial speed of 12 m/s.

- How long does it take for the ball to reach max. height?
- What is the ball's max. height above the release point?
- How long does it take for the ball to reach a point 5.0 m above release point?



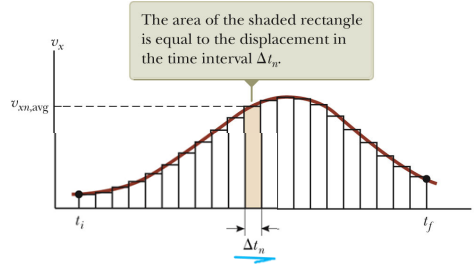
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Kinematic Equations derived using Calculus

- Displacement equals the area under the velocity - time curve

$$\Delta x = \lim_{\Delta t_n \rightarrow 0} \sum_n v_{xn} \Delta t_n = \int_{t_i}^{t_f} v_x(t) dt$$

- The limit of the sum is a definite integral.



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Kinematic Equations - General Calculus Form

- By definition $a = a_{avg} = \frac{\Delta v}{\Delta t} = \frac{dv}{dt}$ As $\Delta t \rightarrow 0$,

$$dv = a dt$$

$$\int_{v_i}^{v_f} dv = \int_{t_i}^{t_f} a_x dt = \int_0^t a_x dt$$

$$v_{x,f} - v_{x,i} = a_x t$$

- Then

$$v_{x,f} = v_{x,i} + a_x t$$

- Similarly:

$$v_{x,avg} = \frac{\Delta x}{\Delta t}$$

$$\text{As } \Delta t \rightarrow 0, v_{avg} \rightarrow v$$

$$\text{so } v_x = \frac{dx}{dt}$$

$$\text{Rearranging } dx = v_x dt$$

$$\int_{x_i}^{x_f} dx = \int_{v_{x,i}}^{v_{x,f}} v_x dt = \int_{v_{x,i}}^{v_{x,f}} (v_{x,i} + a_x t) dt$$

$$x_f - x_i = v_{x,i} t + \frac{1}{2} a_x t^2$$

$$x_f = x_i + v_{x,i} t + \frac{1}{2} a_x t^2$$

Section 2.9

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